

Optimization

4.2 Karush–Kuhn–Tucker (KKT) conditions

José Firmino Aguilar Madeira

From 4.1 to 4.2: what changes?

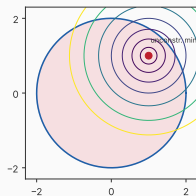
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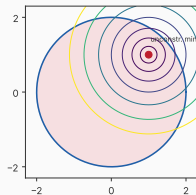
Wide circle, $g = 4 - x_1^2 - x_2^2$. The unconstrained minimum $(1, 1)$ is feasible: $g(\mathbf{x}^*) = 4 - 1 - 1 = 2 > 0$.

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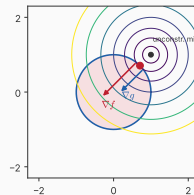
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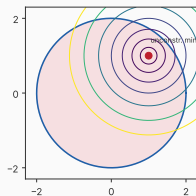
Tight circle, $g = 1 - x_1^2 - x_2^2$. $(1, 1)$ is no longer feasible; $\mathbf{x}^* = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, $g(\mathbf{x}^*) = 0$. **Active**:

$$\nabla f = \lambda \nabla g, \quad 2\left(\frac{1}{\sqrt{2}} - 1\right)(1, 1) = \lambda(-\sqrt{2})(1, 1), \\ \lambda^* = \sqrt{2} - 1 > 0.$$

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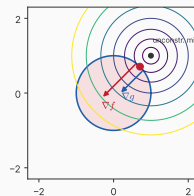
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Same f : **just move the boundary**. When active, it is like an equality that only “pushes” inwards.

The general problem

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \quad \text{s.t.} \quad g_j(\mathbf{x}) \geq 0, \quad j = 1, \dots, J; \quad h_\ell(\mathbf{x}) = 0, \quad \ell = 1, \dots, K.$$

- Course convention (Deb, and 1.1): inequalities in the form $g_j \geq 0$. A constraint $c(\mathbf{x}) \leq b$ is written $g = b - c(\mathbf{x}) \geq 0$.
- **Feasible region** X : all $\mathbf{x}$ that satisfy all the constraints.
- At $\mathbf{x} \in X$, g_j is **active** if $g_j(\mathbf{x}) = 0$ (on the boundary) and **inactive** if $g_j(\mathbf{x}) > 0$. **Active set**: $I(\mathbf{x}) = \{j : g_j(\mathbf{x}) = 0\}$.
- The equalities h_ℓ enter as in 4.1, with multiplier β_ℓ ($= -\lambda_\ell$ of 4.1).

Notes: the bounds $x_i^{(L)} \leq x_i \leq x_i^{(U)}$ are written $x_i - x_i^{(L)} \geq 0$ and $x_i^{(U)} - x_i \geq 0$ and are included in $\mathbf{g}$. Many books and `fmincon` use $c(\mathbf{x}) \leq 0$: the geometry is the same, but the signs are written differently.

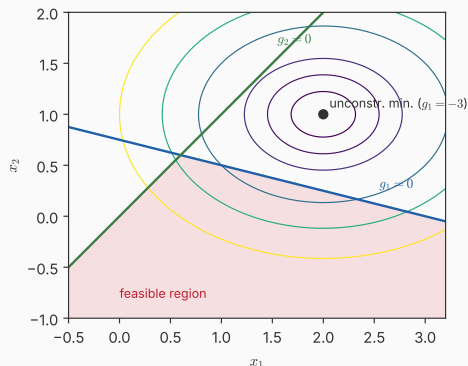
From here on g_j (and $\mathbf{g} = (g_1, \dots, g_J)$) are constraints; the gradient is always written ∇f .

Geometry (1): why does the optimum go to the boundary?

Example 1. $\min (x_1 - 2)^2 + 2(x_2 - 1)^2$ s.t.

$g_1 = 3 - x_1 - 4x_2 \geq 0$, $g_2 = x_1 - x_2 \geq 0$.

- The unconstrained minimum $(2, 1)$ **violates**
 g_1 : $g_1(2, 1) = -3 < 0$.

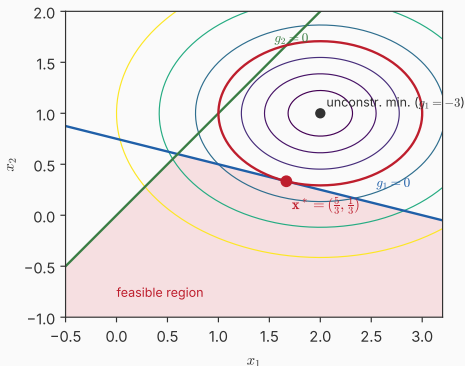


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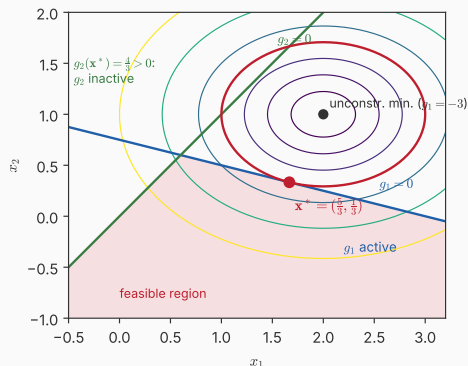
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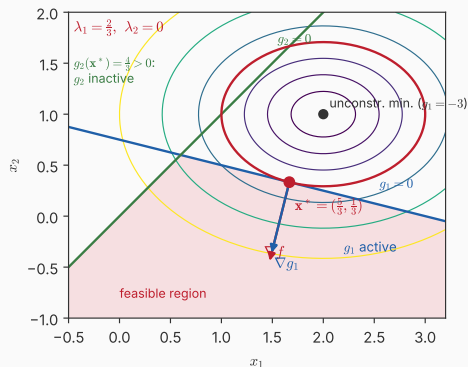
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- At $\mathbf{x}^*$: $g_1 = 0$ — **active**; $g_2 = \frac{4}{3} > 0$ — **inactive**, as if it did not exist.

The same example has both cases: one constraint that “works” and one that does not.



Geometry (2): why $\lambda \geq 0$?

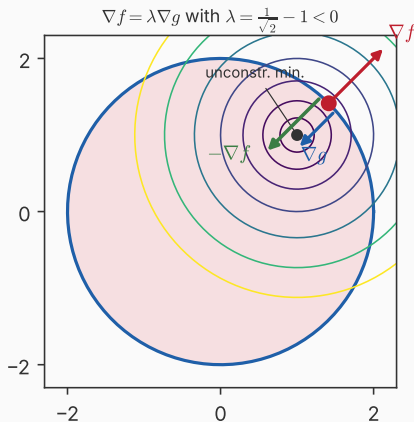
- ∇g_1 points to where g_1 **increases**, that is, to the feasible side $g_1 > 0$ — into X .
- At a boundary optimum, f cannot decrease into X : f increases inwards, so ∇f also points inwards — **the same direction** as ∇g_1 .
- Hence the coefficient relating them is nonnegative:
 $\nabla f(\mathbf{x}^*) = \lambda_1 \nabla g_1(\mathbf{x}^*)$: $\left(-\frac{2}{3}, -\frac{8}{3}\right) = \frac{2}{3}(-1, -4)$,
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- If $\lambda < 0$: on the wide circle of “From 4.1 to 4.2”, the boundary point $(\sqrt{2}, \sqrt{2})$ has $\nabla f = \lambda \nabla g$ with $\lambda = \frac{1}{\sqrt{2}} - 1 < 0$. There ∇f points outwards, $-\nabla f$ inwards — we descend **by entering X** (down to $(1, 1)$): not a minimum.

With the convention $c(\mathbf{x}) \leq 0$, ∇c points outwards and one writes $\nabla f = -\mu \nabla c$, $\mu \geq 0$: the same geometry.



From geometry to the KKT conditions

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(complementary slackness). For each j , at least one of the two is zero. In example 1:
 $\lambda_1 g_1 = \frac{2}{3} \cdot 0 = 0, \lambda_2 g_2 = 0 \cdot \frac{4}{3} = 0$.

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$\min x^2$ s.t. $g = x \geq 0$: $x^* = 0, g(x^*) = 0$ (active), but $2x - \lambda = 0 \Rightarrow \lambda^* = 0$. Inactive $\Rightarrow \lambda = 0$; but $\lambda = 0$ does not imply inactive.

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Putting it all together, at $\mathbf{x}^*$ (with the equalities of 4.1):

$$\text{stationarity } \nabla f = \sum_j \lambda_j \nabla g_j + \sum_\ell \beta_\ell \nabla h_\ell$$

$$\text{feasibility } g_j \geq 0, \quad h_\ell = 0$$

$$\text{complementary slackness } \lambda_j g_j = 0$$

$$\text{sign } \lambda_j \geq 0 \quad (\beta_\ell \text{ free})$$

With $\mathcal{L} = f - \lambda^T \mathbf{g} - \beta^T \mathbf{h}$, the first row is $\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0}$ (the convention used in class).

KKT = Lagrange + sign + complementary slackness.

Without inequalities, it is exactly 4.1.

Karush–Kuhn–Tucker theorem

First-order necessary conditions

If $\mathbf{x}^*$ is a local minimum, with $f, g_j, h_\ell \in C^1$ near $\mathbf{x}^*$, and the gradients of the constraints active at $\mathbf{x}^*$ — $\nabla g_j, j \in I(\mathbf{x}^*)$, and all the ∇h_ℓ — are **linearly independent (LICQ)**, then there exist $\lambda \in \mathbb{R}^J$ and $\beta \in \mathbb{R}^K$ such that

$$\nabla f(\mathbf{x}^*) = \sum_j \lambda_j \nabla g_j(\mathbf{x}^*) + \sum_\ell \beta_\ell \nabla h_\ell(\mathbf{x}^*), \quad g_j(\mathbf{x}^*) \geq 0, \quad h_\ell(\mathbf{x}^*) = 0, \quad \lambda_j g_j(\mathbf{x}^*) = 0, \quad \lambda_j \geq 0.$$

- A point that satisfies the four conditions is called a **KKT point**.
- The LICQ assumption is that of 4.1 extended to the *active* inequalities; without it there may be no multipliers (see below).
- They are **necessary** conditions: they say where to look, they do not guarantee that a minimum has been found (see below).

Solving KKT: recipe and 2^J cases

Recipe

1. Write $\mathcal{L}$ and the conditions. 2. Choose a case: each g_j active ($g_j = 0$) or with $\lambda_j = 0$. 3. Solve the system. 4. Check feasibility and $\lambda_j \geq 0$. 5. Classify the KKT points.

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3	$g_1 = 0, \lambda_2 = 0$	$\mathbf{x} = (\frac{5}{3}, \frac{1}{3}), \lambda_1 = \frac{2}{3}$	$g_2 = \frac{4}{3} > 0, \lambda_1 \geq 0$ ✓
4	$g_1 = g_2 = 0$	$\mathbf{x} = (\frac{3}{5}, \frac{3}{5}), \lambda_1 = \frac{22}{25}, \lambda_2 = -\frac{48}{25}$	$\lambda_2 < 0$ ✗

Case 3 survives — what the figure already showed (g_1 active, g_2 inactive): $\mathbf{x}^* = (\frac{5}{3}, \frac{1}{3}), f^* = 1$. **With J constraints there are 2^J cases:** by hand only for small problems; the algorithms (4.3) avoid this enumeration.

Two active constraints: from *span* to cone

Example 2: checking a candidate. $\min (x_1 - 3)^2 + (x_2 - 2)^2$

s.t. $g_1 = 5 - x_1^2 - x_2^2 \geq 0$, $g_2 = 4 - x_1 - 2x_2 \geq 0$,

$g_3 = x_1 \geq 0$, $g_4 = x_2 \geq 0$.

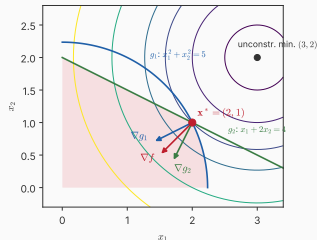
Candidate $\mathbf{x} = (2, 1)$: feasible, active $I = \{1, 2\}$

($g_1 = g_2 = 0$; $g_3 = 2$, $g_4 = 1$), hence $\lambda_3 = \lambda_4 = 0$.

$$\nabla f = (-2, -2), \quad \nabla g_1 = (-4, -2), \quad \nabla g_2 = (-1, -2) :$$

$$\begin{pmatrix} -2 \\ -2 \end{pmatrix} = \lambda_1 \begin{pmatrix} -4 \\ -2 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ -2 \end{pmatrix} \Rightarrow \lambda_1 = \frac{1}{3}, \quad \lambda_2 = \frac{2}{3}.$$

Both ≥ 0 and $\nabla g_1, \nabla g_2$ independent (LICQ): $(2, 1)$ is a KKT point, $f = 2$ — and a global minimum (next page).



From span to cone

4.1: with equalities, ∇f lies in the **space spanned** by the normals, with coefficients of any sign.

4.2: with active inequalities, ∇f lies in the **cone** generated by the active normals: coefficients $\lambda_j \geq 0$.

KKT gives candidates; when are they enough?

In general

local minimum (with LICQ) $\implies$ KKT point

KKT point $\nRightarrow$ minimum

$\min -x^2$ s.t. $g_1 = x + 1 \geq 0$, $g_2 = 2 - x \geq 0$: **three** KKT points.

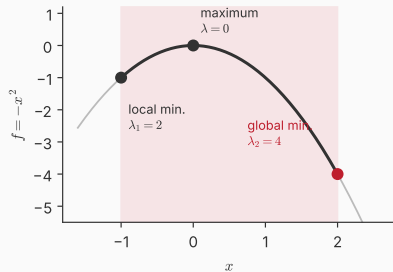
- $x = -1$ (g_1 active, $\lambda_1 = 2$): local minimum, $f = -1$.
- $x = 0$ (both inactive, $\lambda = 0$, $f' = 0$): **maximum**.
- $x = 2$ (g_2 active, $\lambda_2 = 4$): global minimum, $f = -4$.

KKT produces candidates; they must then be classified — as in 4.1.

In the convex case

f convex, g_j concave, h_ℓ affine $\implies$

KKT point $\implies$ global minimum



Why concave g_j ? With $g_j \geq 0$, it is concavity that makes $\{x : g_j(x) \geq 0\}$ convex (with $c \leq 0$ it would be convex c). Examples 1 and 2 are convex: their KKT points are global minima.

Limitations of the KKT conditions

As a solution procedure

They are not an algorithm

They say what a solution must satisfy, not how to find it. Enumerating active sets: up to 2^J cases — only for small problems. The algorithms (4.3) exist to avoid this.

As conditions

No guaranteed minimum

Outside the convex case, a KKT point may be a local minimum, a maximum or a saddle point.

Regularity needed

Without a constraint qualification, a minimum may **fail to be** a KKT point → “When KKT fails”.

Local, with derivatives

In a nonconvex problem there may be a better feasible point elsewhere (Ch. 5). They require derivatives: discrete, nonsmooth or black-box problems need other tools.

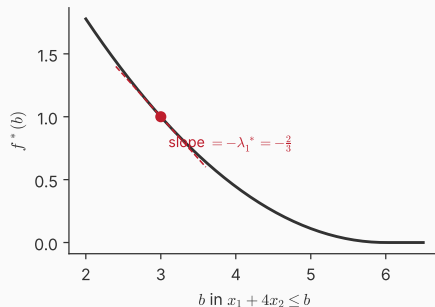
So what are they for? Characterizing solutions, checking candidates, interpreting constraints (multipliers) and underpinning the numerical algorithms.

What the multipliers measure

Relax constraint j : $g_j(\mathbf{x}) + \delta \geq 0$ (the region grows). The optimum varies as

$$\frac{df^*}{d\delta} = -\lambda_j^* \quad (\text{under regularity}).$$

- λ_j^* is the **shadow price** of the constraint: how much is gained per unit of relaxation.
- Inactive: $\lambda_j^* = 0$, relaxing does not help. Active with large λ_j^* : it is the constraint that “costs” most (with constraints on the same scale).
- In example 1: $x_1 + 4x_2 \leq b$, $b = 3 \rightarrow 3.1$: f^* goes from 1 to $\approx 1 - \frac{2}{3} \cdot 0.1 = 0.933$ (exact value: 0.934).
- In the production problem of 1.1 (LP): “an extra hour of machine time is worth 10 €” — the shadow prices of machine time and labour are exactly these λ_j ; it is the information that `linprog` returns in `lambda`.



When KKT fails: LICQ

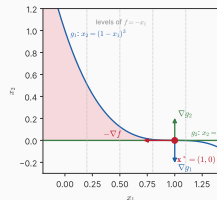
$$\min -x_1 \quad \text{s.t.} \quad g_1 = (1 - x_1)^3 - x_2 \geq 0, \quad g_2 = x_2 \geq 0.$$

- The region is the “wedge” between the cubic and the axis; the minimum is the tip $\mathbf{x}^* = (1, 0)$, $f^* = -1$.
- There both are active: $\nabla g_1 = (0, -1)$, $\nabla g_2 = (0, 1)$ — **dependent** (the cubic reaches the axis with a horizontal tangent).
- $\nabla f = (-1, 0)$ is not a combination of $(0, \mp 1)$: **no** λ_1, λ_2 **exist**. $\mathbf{x}^*$ is a minimum but not a KKT point.

LICQ

Gradients of the equalities and of the **active** inequalities linearly independent at $\mathbf{x}^*$.

Without a constraint qualification, a minimum may **admit no KKT multipliers**.



As in 4.1 (“Why it fails”): dependent normals no longer describe the directions in which one can move.

Note: with all constraints affine (LP, example 1), KKT is necessary without any qualification; there are weaker qualifications than LICQ (MFCQ; Slater, for convex problems).

Exercises

1. $\min -\ln(x_1 + 1) - x_2$ s.t. $3 - 2x_1 - x_2 \geq 0$, $x_1 \geq 0$, $x_2 \geq 0$. $\mathbf{x}^* = (0, 3)$; active: the 1st and $x_1 \geq 0$.
2. $\min x_1^2 + x_2^2$ s.t. $5 - x_1^2 - x_2^2 \geq 0$, $x_1 + 2x_2 = 4$, $x_1, x_2 \geq 0$. $\mathbf{x}^* = (\frac{4}{5}, \frac{8}{5})$, $\beta = \frac{8}{5}$; only the equality is active.
3. Write the KKT conditions for $\min -\mathbf{c}^T \mathbf{x}$ s.t. $\mathbf{b} - \mathbf{A}\mathbf{x} \geq 0$, $\mathbf{x} \geq 0$ (LP) and interpret the multipliers.

$$\mathbf{A}^T \boldsymbol{\lambda} - \mathbf{w} = \mathbf{c}, \boldsymbol{\lambda}, \mathbf{w} \geq 0, \boldsymbol{\lambda}^T (\mathbf{b} - \mathbf{A}\mathbf{x}) = 0, \mathbf{w}^T \mathbf{x} = 0: \text{ it is the dual.}$$

4. For the can of 4.1 (h = height), add $h \leq 6$ cm, that is, $g = 6 - h \geq 0$. Show that the solution changes and compute the new λ .

$$h^* = 6 \text{ cm}, r^* = 4.18 \text{ cm}, A^* = 267.7 \text{ cm}^2; \lambda^* \approx 5.19 \text{ cm}^2/\text{cm}.$$

In MATLAB

`[x,fval,exitflag,output,lambda] = fmincon(f, x0, A, b, Aeq, beq, lb, ub, nonlcon)` solves the problem and returns the multipliers: `lambda.ineqnonlin`, `lambda.eqnonlin`, `lambda.lower`, `lambda.upper`.

THE SAME IN PYTHON

`r = scipy.optimize.minimize(f, x0, method='trust-constr', constraints=[NonlinearConstraint(g, 0, np.inf)]);` the multipliers are in `r.v`.

- **Different sign conventions:** `fmincon` uses $c(\mathbf{x}) \leq 0$ and $c_{eq}(\mathbf{x}) = 0$; SciPy has its own. Inequalities ($c = -g$): `lambda.ineqnonlin` are the λ_j here. Equalities: opposite sign, `lambda.eqnonlin` $= -\beta_\ell$ (`fmincon` adds: $\mathcal{L} = f + \dots + \lambda_{eq}^T c_{eq}$, as in 4.1).
- Rule of thumb: **check against KKT by hand** on a small example (e.g. example 2: $\lambda_1 = \frac{1}{3}$, $\lambda_2 = \frac{2}{3}$) before interpreting.
- By default `fmincon` uses an *interior-point* method, whose idea we will see in 4.3.2.

KEY TAKEAWAYS — 4.2

1. Each inequality, at the optimum, is either **inactive** ($\lambda_j = 0$: as if it did not exist) or **active** (an equality that only pushes inwards: $\lambda_j \geq 0$); in one equation, **complementary slackness** $\lambda_j g_j = 0$ (active with $\lambda_j = 0$ is also possible).
2. **KKT = Lagrange + sign + complementary slackness**: $\nabla f = \sum_j \lambda_j \nabla g_j + \sum_\ell \beta_\ell \nabla h_\ell$; with active inequalities, ∇f lies in the cone of their normals.
3. **Necessary** under a qualification (LICQ; without it, a minimum may have no multipliers); **sufficient in the convex case** (f convex, g_j concave, h_ℓ affine): KKT point $\Rightarrow$ global minimum. Otherwise, candidates to be classified.
4. λ_j is the **shadow price**: how much f^* decreases per unit of relaxation of constraint j .

Next (4.3): KKT says *what* a solution is, not *how* to find it — turn the constrained problem into a sequence of unconstrained problems, solved with the methods of Ch. 3.

References Full list: *Course bibliography*, at the end of the course PDF.

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 12.
- Fletcher, R., *Practical Methods of Optimization*, 2nd ed., Wiley, 1987.
- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — convention $g_j \geq 0$.

Resources

- MATLAB: `fmincon (lambda.ineqnonlin, .eqnonlin, .lower, .upper)`, `linprog` (Optimization Toolbox).
- Python: `scipy.optimize.minimize(..., method='trust-constr', constraints=[NonlinearConstraint(g, 0, np.inf)])`, multipliers in `r.v`.